

Thermodynamic Equilibrium from Reversible One-Quantum Exchange among Discrete Particles

A finite-state counting argument and numerical realization

Matter-as-machine

Abstract

This note studies a minimal model of thermodynamic exchange among discrete particles. A particle is postulated to be a nonempty finite list of quanta and is constrained to remain at size $n \geq 1$. Relative to that minimum size, it has excess occupancy $m = n - 1$. No particular list entry is a persistent baseline: an elementary event may transfer any selected quantum provided that the donor remains nonempty. Each transferred quantum carries energy ε , and total exchange energy is conserved. For M particles and Q excess quanta, and under an explicit occupancy-level coarse-graining, the number of configurations is $\binom{Q+M-1}{Q}$. The uniform microcanonical measure therefore has an exact finite-system marginal distribution and, in the thermodynamic limit at fixed $\nu = \frac{Q}{M}$, a geometric marginal $P(m) = (1-r)r^m$ with $r = \frac{\nu}{1+\nu}$. Defining entropy by $S = k \log \Omega$ and temperature by $\frac{1}{T} = \frac{\partial S}{\partial E}$ gives $r = \exp\left[-\frac{\varepsilon}{kT}\right]$ and $\langle m \rangle = \frac{1}{\left[\exp\left(\frac{\varepsilon}{kT}\right) - 1\right]}$. Thus the exponential thermal parameter is obtained from state counting rather than inserted as a change of variables. For a specified size-independent per-particle transfer kernel, a reversible one-quantum Markov process is shown to have the uniform microcanonical measure as its unique stationary distribution. Time-averaged, multiple-seed simulations of two initially unequal reservoirs agree with the exact finite marginal. The result is a minimal thermodynamics of a fixed-energy exchange ensemble, not a derivation of quantum mechanics, a full Bose gas, or the Planck spectrum.

1. Purpose and scope

The central question is deliberately narrow:

Can temperature and a Bose-Einstein-shaped mean occupancy arise from a concrete one-quantum exchange process among discrete particles, without first imposing the substitution $r = \exp\left[-\frac{\varepsilon}{kT}\right]$?

The answer is yes for the idealized model defined below. The result has two parts:

- a counting result that determines the equilibrium distribution and its temperature parameter; and
- a stochastic dynamics that actually approaches that equilibrium by reversible one-quantum transfers.

The contribution is a deliberately simpler route to familiar equilibrium relations: standard thermodynamic quantities and a Bose-Einstein-shaped single-energy occupancy arise from discrete reversible matter exchange, without assuming an exponential thermal factor or the machinery of quantum mechanics. The coarse-graining and kinetic kernel required for that result are stated explicitly below.

Ontological postulate. Matter is discrete. A particle is identified with a finite nonempty list of elementary instructions, not merely represented by such a list for mathematical convenience. The empty list represents absence of a particle. Consequently, an existing particle has total length $n \geq 1$. Writing $n = 1 + m$, the exchangeable occupancy satisfies $m \geq 0$. The condition $n \geq 1$ is a boundary on total list length; it does not label any particular instruction as a protected baseline.

Thermodynamic coarse-graining postulate. Instruction identity, direction, and position in a list may matter to other dynamics, but they are not resolved by the present equilibrium model. Its energy and thermodynamic state depend only on the occupancies $(m_1, \dots, m_M)$. Occupancy configurations are assigned equal weight. Equivalently, any unresolved list-level degeneracy is assumed to give the same multiplicative factor to every fixed- Q occupancy configuration. This postulate is essential: if list order or instruction identity gives occupancy-dependent multiplicities, the stars-and-bars measure and the results below need not follow.

This paper establishes consequences of that postulate only within the stated reversible exchange model. It does not establish that real elementary particles possess this structure.

Mathematically, the counting is the familiar distribution of indistinguishable quanta among distinguishable containers and is closely related to standard Bose-Einstein occupancy counting. At the level of stochastic occupancy dynamics, the model also overlaps with zero-range and related particle-transfer processes [6]. The claim is therefore not that the combinatorial formula or the general transfer-process mathematics is new. The distinct claim is interpretive: the occupancies are proposed as the literal discrete internal matter of particles, not as site occupancies in an auxiliary stochastic model. Scientific support for that ontology requires distinctive empirical predictions beyond reproducing established equilibrium statistics.

Several topics are intentionally excluded. There is no spatial escape in the equilibrium derivation, no cosmological loss or “tired light,” no creation of matter, and no pressure-volume law. These belong to separate models and should not be used to justify the equilibrium result proved here.

2. Discrete particles and conserved exchange energy

Consider $M \geq 2$ distinguishable particles. Particle i has total list length

$$n_i = 1 + m_i, \quad m_i \in \{0, 1, 2, \dots\}.$$

The constraint $n_i \geq 1$ ensures that a particle always exists. The variable $m_i = n_i - 1$ is a count relative to this minimum, not a label attached to a protected instruction. Whenever $n_i > 1$, any selected instruction may be transferred while leaving at least one instruction behind. Each transferred instruction carries the same energy $\varepsilon > 0$. Hence

$$E_i = \varepsilon m_i, \quad E = \sum_{i=1}^M E_i = \varepsilon Q, \quad Q = \sum_{i=1}^M m_i.$$

An elementary exchange transfers one excess instruction:

$$(m_i, m_j) \rightarrow (m_i - 1, m_j + 1), \quad m_i \geq 1.$$

The reverse event is also allowed. Every event conserves Q and therefore E . No instruction is created or destroyed in this closed model.

The minimum-size subtraction is an energy convention, not a persistent physical quantum. If the total list energy is instead defined as εn_i , every particle receives the same additive offset ε . The total energy then changes by $M\varepsilon$, but exchange probabilities, entropy differences, temperature, and the distribution of m_i are unchanged.

3. Exact finite-system state counting

3.1. Number of configurations

A microconfiguration at the occupancy level is an ordered M -tuple

$$(m_1, \dots, m_M), \quad m_i \geq 0, \quad \sum_i m_i = Q.$$

The number of such weak compositions is the stars-and-bars count

$$\Omega(Q, M) = \binom{Q + M - 1}{Q} = \binom{Q + M - 1}{M - 1}.$$

The coarse-graining postulate assigns all these occupancy configurations equal weight. Section 6 supplies an explicit occupancy-level Markov process for which the corresponding uniform stationary probability follows from detailed balance and ergodicity. It does not derive that coarse-graining from finer list-level dynamics.

3.2. Exact marginal of one particle

Fix $m_1 = m$. The remaining $Q - m$ quanta may be distributed among $M - 1$ particles in

$$\Omega(Q - m, M - 1) = \binom{Q - m + M - 2}{Q - m}$$

ways. Therefore the exact finite-system marginal is

$$P_{Q,M}(m) = \frac{\binom{Q - m + M - 2}{Q - m}}{\binom{Q + M - 1}{Q}}, \quad 0 \leq m \leq Q.$$

Its adjacent ratio is

$$\frac{P_{Q,M}(m + 1)}{P_{Q,M}(m)} = \frac{Q - m}{Q - m + M - 2}.$$

This ratio depends weakly on m at finite Q and M . Consequently the exact finite marginal is not exactly geometric. That distinction matters: the geometric law appears as a controlled thermodynamic-limit result, not as an identity for every finite ensemble.

3.3. Thermodynamic limit

Let $Q, M \rightarrow \infty$ while the mean excess occupancy

$$\nu = \frac{Q}{M}$$

remains fixed. For each fixed m ,

$$\frac{Q-m}{Q-m+M-2} \rightarrow \frac{Q}{Q+M} = \frac{\nu}{1+\nu} = r.$$

Normalization then gives

$$P(m) = (1-r)r^m, \quad m \geq 0, \quad r = \frac{\nu}{1+\nu}.$$

The mean of this distribution is

$$\langle m \rangle = \frac{r}{1-r} = \nu.$$

Thus the geometric law is both the limiting marginal of uniform compositions and the distribution whose mean equals the conserved excess occupancy per particle.

4. Entropy and the emergence of temperature

Define microcanonical entropy by

$$S(E, M) = k \log \Omega(Q, M), \quad Q = \frac{E}{\varepsilon},$$

where k fixes the temperature unit. Because energy is discrete, define the finite-system inverse temperature by the entropy gained when one quantum is added:

$$\frac{1}{T_{Q,M}} := \frac{S(Q+1, M) - S(Q, M)}{\varepsilon} = \left(\frac{k}{\varepsilon}\right) \log\left(\frac{Q+M}{Q+1}\right).$$

This is exact. In the thermodynamic limit it approaches

$$\frac{1}{T} = \left(\frac{k}{\varepsilon}\right) \log\left(\frac{Q+M}{Q}\right) = \left(\frac{k}{\varepsilon}\right) \log\left(1 + \frac{1}{\nu}\right).$$

The same limiting expression follows from the usual derivative. Using Stirling's approximation at large Q and M gives

$$\frac{S}{k} \approx (Q+M) \log(Q+M) - Q \log Q - M \log M.$$

In terms of $\nu = \frac{Q}{M}$,

$$S \approx kM[(1+\nu) \log(1+\nu) - \nu \log \nu].$$

For the smooth limiting entropy, temperature is equivalently

$$\frac{1}{T} = \frac{\partial S}{(\partial E)_M}.$$

Applying this derivative and using $E = \varepsilon Q$ reproduces

$$\frac{1}{T} \approx \left(\frac{k}{\varepsilon}\right) \log\left(\frac{Q+M}{Q}\right) = \left(\frac{k}{\varepsilon}\right) \log\left(1 + \frac{1}{\nu}\right).$$

Equivalently, in that same limit,

$$\frac{\varepsilon}{kT} \approx \log\left(1 + \frac{1}{\nu}\right).$$

Solving the limiting relation for ν yields

$$\nu = \langle m \rangle = \frac{1}{\exp\left(\frac{\varepsilon}{kT}\right) - 1}.$$

The limiting geometric ratio becomes

$$r = \frac{\nu}{1+\nu} = \exp\left[-\frac{\varepsilon}{kT}\right].$$

The full limiting marginal is therefore

$$P(m) = \left(1 - \exp\left[-\frac{\varepsilon}{kT}\right]\right) \exp\left[-m \frac{\varepsilon}{kT}\right].$$

The exponential factor has not been postulated. It follows from the logarithmic growth of the number of configurations and the thermodynamic definition

$$\frac{1}{T} = \frac{\partial S}{\partial E}.$$

This is the mathematical single-energy Bose-Einstein occupancy form. It is not yet the full distribution of radiation over frequency. A Planck spectrum also requires an energy label such as $\varepsilon = \hbar\omega$, a density of states, and the appropriate photon interpretation.

5. Thermodynamic relations of the ideal exchange ensemble

The relations in this section use the thermodynamic-limit temperature obtained in Section 4. They are not asserted as exact finite- M, Q identities.

5.1. Internal energy

The mean exchange energy per particle is

$$u(T) = \varepsilon \langle m \rangle = \frac{\varepsilon}{\exp\left(\frac{\varepsilon}{kT}\right) - 1}.$$

For M particles,

$$U(T, M) = M \frac{\varepsilon}{\exp\left(\frac{\varepsilon}{kT}\right) - 1}.$$

Any baseline energy is additive and omitted here.

5.2. Heat capacity

Let $x = \frac{\varepsilon}{kT}$. Differentiation gives the heat capacity per particle

$$c(T) = \frac{du}{dT} = kx^2 \frac{\exp(x)}{(\exp(x) - 1)^2} > 0.$$

At high temperature, $x \ll 1$ and

$$u(T) \approx kT - \frac{\varepsilon}{2} + O(T^{-1}), \quad c(T) \rightarrow k.$$

At low temperature,

$$u(T) \approx \varepsilon \exp\left[-\frac{\varepsilon}{kT}\right], \quad c(T) \rightarrow 0.$$

5.3. Zeroth law: an operational temperature

Place two subsystems A and B in weak exchange contact while conserving $E_A + E_B$. The total multiplicity is

$$\Omega_{\text{tot}(E_A)} = \Omega_{A(E_A)} \Omega_{B(E-E_A)},$$

and the most probable division maximizes $S_{\text{tot}} = S_A + S_B$. At the maximum,

$$\frac{\partial S_A}{\partial E_A} = \frac{\partial S_B}{\partial E_B},$$

so

$$\frac{1}{T_A} = \frac{1}{T_B}.$$

Within the stated symmetric, connected exchange model, two bodies have the same temperature when their late-time mean energy flux vanishes. Outside this model, vanishing flux alone is not sufficient: kinetic blocking or disconnected exchange channels can also prevent a flux between unequal states. This criterion is stronger than identifying temperature with an interaction frequency: changing the event rate changes how quickly equilibrium is reached, whereas changing $\frac{Q}{M}$ changes which equilibrium is reached.

5.4. First law in the closed exchange model

Every transfer removes one quantum from one particle and gives it to another. Therefore

$$\Delta E_A + \Delta E_B = 0.$$

There is heat exchange but no work term in the present model. Its first-law content is simply conservation of exchange energy.

5.5. Equilibrium concentration and typicality

Starting far from equilibrium, overwhelmingly more configurations correspond to energy divisions near the maximum of $S_A + S_B$ than to extreme divisions. The irreducible exchange process of Section 6, whose stationary measure is uniform, therefore approaches the high-multiplicity region with high probability. Microscopic trajectories can fluctuate away from it; the statistical claim is that the equilibrium macrostate dominates for large systems. This is an equilibrium concentration statement, not a proof that entropy increases monotonically along every stochastic trajectory.

The model also has a simple low-temperature limit. As $T \rightarrow 0$, $\nu \rightarrow 0$, all excess occupancies vanish, and the unique occupancy configuration is $m_i = 0$ for every particle. Hence $S \rightarrow 0$ within this idealized occupancy description.

6. Reversible one-quantum exchange dynamics

The counting above describes equilibrium. We now give a process that reaches it. At each discrete update:

1. Choose an ordered pair of distinct particles (i, j) uniformly.
2. If $m_i > 0$, transfer one quantum from i to j .
3. If $m_i = 0$, perform no transfer.

Uniform selection of the donor **particle**, rather than uniform selection among all transferable instructions, is an independent kinetic postulate of this model. It makes the attempted transfer rate size-independent. Discreteness and reversibility alone do not select this kernel: choosing instructions uniformly would make donor rates occupancy-dependent and can produce a different stationary measure. A physical application must therefore justify or test the size-independent per-particle interaction rate.

For any two neighboring configurations a and b connected by moving one quantum from i to j ,

$$P(a \rightarrow b) = \frac{1}{[M(M-1)]} = P(b \rightarrow a).$$

Thus the transition matrix is symmetric on every allowed edge. The uniform measure

$$\pi(a) = \frac{1}{\Omega(Q, M)}$$

satisfies detailed balance:

$$\pi(a)P(a \rightarrow b) = \pi(b)P(b \rightarrow a).$$

The state graph is connected because quanta can be transferred one at a time between any particles. The self-loops at empty donors make the chain aperiodic. The finite chain is therefore irreducible and aperiodic, so the uniform microcanonical distribution is its unique stationary law and the process converges to it from every initial configuration.

This argument identifies the essential assumptions:

- one-quantum transfers;
- conservation of Q ;
- reversibility with equal forward and reverse edge weights; and
- sufficient connectivity to explore the full fixed- Q state space.

The first item alone is not sufficient; the equal edge weights generated by uniform particle-pair selection are what make the stationary measure uniform.

If physical exchange rules violate these conditions, a different stationary law may result.

7. Two-reservoir numerical experiment

7.1. Reversible exchange model

The implementation `two_bath_reversible_exchange.cpp` uses two reservoirs with 1000 particles each. Reservoir A begins with mean excess occupancy 8 and reservoir B with mean excess occupancy 1. Thus the conserved global mean is

$$\nu_{\text{global}} = \frac{8 + 1}{2} = 4.5.$$

Eight runs use consecutive fixed seeds. Each run has 5 million burn-in updates followed by 20 million measured updates; the full 2000-particle histogram is sampled every 100,000 measured updates. The resulting time-averaged values, reported as mean $\pm$ sample standard deviation across runs, are

quantity	mean	run SD	target
ν_A	4.50719	0.00926	4.50000
ν_B	4.49282	0.00926	4.50000
net cross-flux	-4×10^{-6}	1.7×10^{-5}	0

The equilibrium parameters predicted from the conserved global mean are

$$r = \frac{4.5}{5.5} = 0.818182, \quad \beta\varepsilon = \log\left(1 + \frac{1}{4.5}\right) = 0.200671.$$

The reservoir means fluctuate around 4.5 rather than remaining exactly equal, as expected for finite systems.

7.2. Exact finite-law test

The code constructs the exact finite marginal $P_{9000,2000}(m)$ from its adjacent ratio, without fitting a parameter to either reservoir. It then compares that prediction with the pooled post-burn-in histogram. Across the eight runs,

$$\text{TV}(\text{exact}) = 0.002702 \pm 0.000186.$$

For reference, comparison with the thermodynamic-limit geometric law fixed by the conserved global mean $\nu = 4.5$ gives

$$\text{TV}(\text{geometric}) = 0.002715 \pm 0.000162.$$

To quantify temporal dependence, the code estimates the integrated autocorrelation time of the ν_A snapshot series by summing positive empirical autocorrelations. Across runs,

$$\tau_{\text{int}} = 1.108 \pm 0.235 \text{ snapshots}, \quad \text{ESS} = 92.98 \pm 14.75$$

out of 200 stored snapshots per run. These spreads are sample standard deviations across eight runs, not confidence intervals. Particle values within a snapshot are also constrained by fixed Q , so these diagnostics do not prove that the TV discrepancy is mostly sampling noise. They show close finite-run agreement with the exact law at the tested scale.

8. Relation to Bose-Einstein statistics

The limiting marginal

$$P(m) = (1 - \exp[-\beta\varepsilon]) \exp[-\beta\varepsilon m]$$

and mean

$$\langle m \rangle = \frac{1}{\exp(\beta\varepsilon) - 1}$$

are the standard single-energy bosonic occupancy formulas at zero chemical potential. Bose’s original radiation counting and Einstein’s extension to an ideal gas established the historical quantum-statistical setting for these expressions [1, 2].

The present construction differs in interpretation: the containers are identified with finite nonempty discrete particles and the quanta with transferable list entries. The boundary of the result—single energy rather than a spectrum, occupancy mathematics rather than quantum mechanics, and closed equilibrium rather than formation or escape—is collected in Section 9.

9. What has and has not been derived

Within the stated model, the following results are derived:

- the exact finite marginal $P_{Q,M}(m)$;
- its geometric thermodynamic limit;
- entropy $S = k \log \Omega$;
- the thermodynamic-limit temperature relation $r = \exp[-\frac{\varepsilon}{kT}]$;
- the Bose-Einstein-shaped mean occupancy;
- internal energy and heat capacity;
- equality of temperature as the condition for maximal total entropy and zero late-time mean flux; and
- convergence of a specified reversible exchange chain to the microcanonical equilibrium.

The following are not derived:

- the microscopic origin or numerical value of ε ;
- the physical identity of the transferable quanta;
- the Planck spectral factor or density of states;
- pressure, volume, mechanical work, or an equation of state;
- chemical potential when particle number changes;
- creation, annihilation, reset events, or cosmological redshift; and
- quantum mechanics or the ontology of real elementary particles.

This boundary is not cosmetic. It separates a reproducible mathematical result from hypotheses that require additional dynamics and empirical tests.

10. Reproducibility

The simulations are deterministic given their pseudorandom seeds and are implemented in the accompanying source files:

- `two_bath_reversible_exchange.cpp`: symmetric reversible exchange between two initially unequal reservoirs.

The reproducibility archive is the public repository github.com/matterasmachine/matterasmachine. The corresponding project folder is available here. The release of this manuscript must be tagged or archived together with that source so that the reported code and document resolve to the same immutable revision; no DOI has yet been assigned.

The simulation is not an independent proof of equilibrium: detailed balance already supplies that proof. It validates the implementation, demonstrates convergence from unequal reservoirs, and measures finite-run agreement with the exact marginal. The reported run-to-run standard deviations and ESS diagnostics are not confidence intervals or precision estimates of universal constants.

11. Conclusion

A minimal reversible exchange process among nonempty discrete particles, together with the stated coarse-graining and size-independent particle-pair kernel, produces a coherent ideal thermodynamics. Uniform fixed-energy configurations have an exact finite marginal that approaches a geometric law. The logarithm of their multiplicity supplies entropy; differentiating entropy with respect to energy supplies temperature; and the resulting thermodynamic-limit geometric ratio is $r = \exp[-\frac{\epsilon}{kT}]$. A symmetric one-quantum Markov process converges to this equilibrium, while two-reservoir simulations show energy equalization and vanishing late-time net flux.

The main conceptual result is therefore modest but concrete:

Within the stated occupancy coarse-graining and exchange kernel, for discrete particles that reversibly exchange conserved equal-energy quanta, temperature measures the equilibrium energy-per-particle scale selected by the number of accessible configurations. The Bose-Einstein denominator is the resulting occupancy relation, not an independently imposed exponential rule.

12. References

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Appendix A: finite marginal and geometric convergence

For completeness, expand the exact marginal as a product. From the adjacent ratio,

$$P_{Q,M}(m) = P_{Q,M}(0) \prod_{j=0}^{m-1} \frac{Q-j}{Q-j+M-2}.$$

Also,

$$P_{Q,M}(0) = \frac{M-1}{Q+M-1}.$$

At fixed $\nu = \frac{Q}{M}$ and fixed m ,

$$P_{Q,M}(0) \rightarrow \frac{1}{1+\nu} = 1-r$$

and every factor in the finite product approaches

$$\frac{Q}{Q+M} = \frac{\nu}{1+\nu} = r.$$

Therefore

$$P_{Q,M}(m) \rightarrow (1-r)r^m.$$

The restriction to fixed m is the ordinary pointwise thermodynamic-limit statement. Uniform error bounds over a range of m require specifying how that range grows with M .